TEXORA · LECTURE SHEET

Weaving of Synthetic and Jute Fabrics

Two extremes of the weaving world. Synthetic filament asks for gentleness, static control and heat discipline; jute asks for mass, moisture and machinery built like a press. This sheet studies both weavings end to end — preparation, loom, settings, defects — with the fabric arithmetic of each worked in its own counting system.

FM-4101 · ADVANCED WEAVING TECHNOLOGY · TOPIC 3 · SPECIALIZATION — FABRIC MANUFACTURING

1. Synthetic filament weaving — the material's terms

Polyester and nylon filaments arrive smooth, strong, uniform — and with four weaving liabilities: **low moisture regain** (static charge at every guide), **low inter-filament friction** (packages slough, loops slip, selvedges draw in), **heat sensitivity** (friction hotspots glaze or melt filaments), and **elastic memory** (tension variations print themselves into the cloth as barré and tight picks). The whole preparation and loom setup is an answer to those four words.

Preparation

- **Sizing:** filament warps are sized for abrasion protection and filament binding, not strength — water-soluble PVA or acrylic sizes at low add-on, applied on filament sizing machines at low tension; unsized (water-jet) warps rely on interlaced (air-entangled) yarn where the process allows.
- **Count language:** filament is counted in denier: 150 denier = 150/9 = **16.7 tex**; 75 denier = **8.3 tex** — convert before any Sheet 5 calculation, or the GSM will be wrong by the factor nobody notices.
- **Static and humidity:** anti-static agents in the size, humidified sheds, earthed guides; a filament warp in a dry room is a charge generator with a beam attached.

On the loom

- **Machine choice:** water-jet in its kingdom (Sheet 2), air-jet and rapier elsewhere; low, even warp tension and a small, late shed to cut filament abrasion; ceramic or polished contact surfaces everywhere — one burred guide writes a line of broken filaments down the whole warp.
- **Typical rate, calculated:** water-jet at 1,000 ppm, 90 PPI, 85% efficiency: production = 1,000 × 60 × 0.85 ÷ 90 ÷ 39.37 = **14.4 m/h** — fine cloth at coarse-cloth speeds, which is the filament bargain.
- **Defects to design against:** broken filaments (fuzz balls, shiners where a bundle reflects light), looped weft from slack insertion, reed marks amplified by smooth yarns, and oil spots that dyeing

will frame forever.

> **WORKED EXAMPLE — A POLYESTER FABRIC BY SHEET 5'S METHOD**
>
> Warp 150 denier (16.7 tex) at EPI 120, crimp 6%; weft 100 denier (11.1 tex) at PPI 100, crimp 4%.
>
> Warp = 120 × 39.37 × 1.06 × 16.7 ÷ 1,000 = **83.6 g/m²**.
>
> Weft = 100 × 39.37 × 1.04 × 11.1 ÷ 1,000 = **45.4 g/m²**.
>
> **GSM = 129 g/m²** — taffeta/lining territory, reached with the same formula as cotton: only the counting language changed.

2. Jute weaving — the material's terms

Jute inverts every synthetic term: coarse counts, hairy and irregular yarns, low elongation, high bending stiffness, and a fibre that weaves best damp. Yarn is counted in the jute system — **pounds per spindle**, the weight of 14,400 yards: 1 lb/spindle = 34.45 tex, so a warp of 8 lb/spindle is **275.6 tex** and a weft of 9 lb/spindle is **310.1 tex** (the conversion YM-4201 derived from the same definition).

Preparation

- **Batching and softening:** jute is oiled and emulsified (batching oil + water emulsion), then binned to soften — weaving jute dry is weaving wire.
- **Winding and warping:** heavy packages, large creels, low speeds; beaming with strong, even tension because the loom will not correct a slack sheet.
- **Sizing:** light or none for sacking; hessian warps may take a light size — the counts are so coarse that add-on percentages translate to large absolute weights.

On the loom

- **Machine:** the heavy jute loom — massive frame, strong picking and beat-up, broad temples, cop or large-pirn shuttle (jute remains shuttle territory in much of its industry) or heavy projectile/rapier in modern mills; wide healds and reeds with generous spacing for hairy yarns.
- **Humidity:** the jute shed is kept humid and the warp may be damped — moisture is a process input, as with sizing in cotton.
- **Products:** hessian (lighter, finer — bags, wrapping, backing), sacking (heavier, coarser), carpet backing, geotextiles — graded by weight and construction, not fashion.

> **WORKED EXAMPLE — A HESSIAN CLOTH**
>
> EPI 12, PPI 13; warp 8 lb/spindle (275.6 tex), crimp 3%; weft 9 lb/spindle (310.1 tex), crimp 2%.
>
> Warp = 12 × 39.37 × 1.03 × 275.6 ÷ 1,000 = **134 g/m²**.

Weft = 13 × 39.37 × 1.02 × 310.1 ÷ 1,000 = **162 g/m²**.

GSM = 296 g/m² — twice the cotton reference cloth of FM-3207, at one-eighth of its thread density: in jute, the count does what the density does elsewhere.

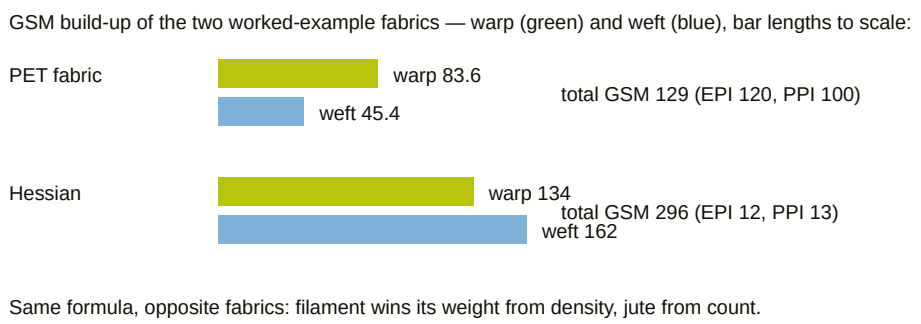


Figure 1 — Where the weight sits in a filament fabric and a jute fabric: warp/weft split of the two worked examples.

3. The two weavings side by side

ASPECT	SYNTHETIC FILAMENT	JUTE
Counts	Denier/tex, fine (8–17 tex typical)	lb/spindle, coarse (275–310 tex)
Preparation key	Film-forming size, anti-static, low tension	Oiling/emulsion, binning, moisture
Loom	Water-jet / air-jet / rapier, light build	Heavy jute loom, shuttle or heavy shuttleless
Tension policy	Low and very even	High and even
Signature defects	Broken filaments, shiners, barré	Hairiness faults, reediness, weight variation
Shed climate	Humidified against static	Humidified for fibre pliability

The symmetry is instructive: both weavings fail when the material's terms are ignored, and both are routine when preparation obeys them. Weaving technology, at this end of the course, is the discipline of letting each fibre state its terms first.

Glossary

Denier — grams per 9,000 m; filament count (denier ÷ 9 = tex). **lb/spindle** — jute count: weight in lb of 14,400 yd (1 lb/spindle = 34.45 tex). **Batching** — oiling and conditioning jute before spinning/weaving preparation. **Hessian** — plain jute cloth of lighter construction; **sacking** the heavier grade. **Shiner** — lustrous filament fault patch. **Interlacing** — air-entangling filament yarn to bind filaments without size.

Key points

• Synthetic terms: static, slip, heat, memory — answered by size chemistry, humidity, low even tension, clean surfaces (rate worked: 14.4 m/h).
• Convert before calculating: 150 den = 16.7 tex; the PET example builds to 129 GSM by the standard formula.
• Jute terms: coarse, hairy, stiff, damp — lb/spindle counts (8 lb = 275.6 tex), emulsion preparation, heavy looms; hessian example = 296 GSM.
• Filament gets weight from density; jute gets it from count — one formula, two strategies.

Practice

1. Convert 300 denier polyester to tex, and find the GSM contribution of a warp of it at EPI 90 with 5% crimp.
2. A sacking cloth uses warp 10 lb/spindle and weft 11 lb/spindle at EPI 10, PPI 9, crimps 3% and 2%. Estimate its GSM.
3. Why does raising shed humidity help both weavings, for opposite reasons?

Answer key

1. 300 ÷ 9 = **33.3 tex**; contribution = 90 × 39.37 × 1.05 × 33.3 ÷ 1,000 = **123.9 g/m²**.
2. Warp tex = 344.5, weft = 379.0. Warp = 10 × 39.37 × 1.03 × 344.5 ÷ 1,000 = 139.7; weft = 9 × 39.37 × 1.02 × 379.0 ÷ 1,000 = 137.0; GSM ≈ **277 g/m²**.
3. Filament: moisture bleeds away static charge, so yarns stop repelling, clinging and balling. Jute: moisture plasticises the stiff fibre, raising pliability and cutting breakages. Same lever, different physics — which is why "humidify" is never a complete instruction without its reason.